CFSS
CYBER & FORENSIC SECURITY SOLUTIONS

INTERNSHIP PROJECT REPORT

ON

SOC OPERATIONS & THREAT DETECTION MASTERY

USING WAZUH SIEM

A Practical Implementation of Security Monitoring, Log Analysis & Threat Detection in a SOC Environment

INTERNSHIP PROVIDER

CFSS Cyber & Forensic Security Solutions
(Cyber Security & Forensics Training, Internships & Placement Company)

SUBMITTED BY

Name	:	Parag Patel
Roll No.	:	CFSS/Intern/2026
Course	:	SOC Operations & Threat Detection Mastery
Batch	:	Summer Global Internship 2026
Date	:	24 June 2026

UNDER THE GUIDANCE OF

Mentors & Trainers
CFSS Cyber & Forensic Security Solutions

MONITOR SECURITY EVENTS | DETECT THREATS | ALERT IN REAL TIME | RESPOND EFFECTIVELY

OFFER LETTER

CFSS

Ref Noas/cfss2026/b1/q7mm
Date: 05/06/2026

Dear Patel Parag Dasharathbhai

We are pleased to inform you that you have been selected to participate in the **CFSS GLOBAL SUMMER INTERNSHIP 2026** offered by the **CFSS Digital Intelligence Pvt. Ltd.** This program is designed to provide project-based learning opportunities in the field of cybersecurity and forensics.

Details of the Internship Program:
- Duration: **01 Month**
- Domain: **SOC Analyst**
- Starting Date: **05th June 2026**
- End Date: **05th July 2026**
- Mode: **Remote & Project-Based**

Important Notes:
- This is a project-based internship, and no formal training sessions will be provided.
- Participants are expected to complete assigned tasks and projects independently under the guidance of their respective mentors.

Perks and Benefits:
- Official Offer Letter upon acceptance of this internship.
- Completion Certificate upon successful fulfillment of the internship requirements.
- Letter of Recommendation (LOR) based on your performance during the program.

Please confirm your acceptance of this offer by replying to this email or contacting us at the earliest. Upon confirmation, further details regarding project assignments and mentor allocation will be shared with you.

For any queries, please feel free to contact us at:
- Phone: +91 99738 72108
- Email: officialcfss@gmail.com

We look forward to your valuable contributions to this program and wish you success in your cybersecurity journey.

Yours sincerely,

Internship Manager,
CFSS Digital Intelligence Pvt. Ltd.
ISO 9001 | IAF Accredited | MSME Registered

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **CFSS (Cyber Security & Forensics Services)** for providing me with the opportunity to participate in the **SOC Operations & Threat Detection Mastery Internship Program**.

This internship provided valuable exposure to Security Operations Center (SOC) operations, threat detection methodologies, log analysis techniques, incident response practices, and cybersecurity monitoring workflows. Through this project, I gained practical experience in deploying and managing a Security Information and Event Management (SIEM) environment, analysing security logs, monitoring alerts, and investigating security events.

I would also like to thank the mentors, coordinators, and the entire CFSS team for their guidance, support, and commitment to delivering a practical learning experience. Their expertise and encouragement played a significant role in strengthening my technical knowledge and analytical skills.

The knowledge and practical experience gained during this internship enhanced my understanding of cybersecurity operations, digital forensics concepts, threat hunting methodologies, incident detection strategies, and security monitoring techniques.

Finally, I would like to thank everyone who supported and motivated me throughout this learning journey and encouraged me to continuously improve my cybersecurity skills.

Parag Patel
SOC Analyst Intern
CFSS Global Internship Program 2026

EXECUTIVE SUMMARY

The **SOC Operations & Threat Detection Mastery Project** was carried out as part of the **CFSS Summer Global Internship 2026**. The primary objective of this project was to design, deploy, and operate a Security Operations Center (SOC) monitoring environment using the **Wazuh Security Information and Event Management (SIEM)** platform.

In today's rapidly evolving cyber threat landscape, organizations require continuous monitoring and effective threat detection mechanisms to identify and respond to malicious activities. This project focuses on building a practical SOC laboratory environment capable of collecting, analyzing, and monitoring security events generated from Windows endpoints.

The project involved the deployment of a Wazuh Manager and Wazuh Agent, enabling centralized log collection and security monitoring. Various security events, including successful logins, failed authentication attempts, command executions, and simulated attack activities, were monitored and analysed to understand how modern SOC teams identify potential threats.

Several threat detection scenarios were performed during the project, including brute-force attack simulations, malware detection using the EICAR test file, command execution monitoring, and event log analysis. Alerts generated by the SIEM platform were investigated to understand detection logic, security rules, and incident response workflows.

Furthermore, customized dashboards were developed to visualize security alerts, agent status, and rule-based detections. These dashboards provide valuable insights into the security posture of the monitored environment and demonstrate how SOC analysts use security telemetry to identify suspicious activities.

The successful completion of this project enhanced practical knowledge in SIEM technologies, threat detection methodologies, security monitoring, incident investigation, log analysis, and cybersecurity operations. The experience gained through this project contributes significantly toward developing professional skills required for SOC Analyst, Threat Hunter, and Digital Forensics roles.

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INTRODUCTION

Cybersecurity has become a critical requirement for organizations due to the increasing number of cyber threats, malware attacks, unauthorized access attempts, and data breaches. To effectively detect and respond to these threats, organizations rely on Security Operations Centers (SOCs) and Security Information and Event Management (SIEM) platforms.

This project focuses on building a practical SOC environment using **Wazuh SIEM** for security monitoring and threat detection. The environment collects and analyses security logs generated by monitored endpoints to identify suspicious activities and potential security incidents.

The project includes endpoint monitoring, log analysis, event investigation, brute-force attack detection, malware detection using the EICAR test file, and command execution monitoring. Various security alerts generated by Wazuh are analysed to understand real-world incident detection and response workflows.

In addition, dashboards are developed to visualize security events, alert trends, and endpoint status. This project provides practical experience in SIEM operations, threat detection, incident response, threat hunting, and cybersecurity monitoring while strengthening the skills required for a SOC Analyst role.

PROJECT OBJECTIVES

The primary objective of this project is to develop a practical understanding of Security Operations Center (SOC) activities using the Wazuh SIEM platform. The project aims to simulate a real-world security monitoring environment and gain hands-on experience in threat detection, log analysis, and incident investigation.

The specific objectives of this project are:

- To deploy and configure a Wazuh SIEM environment for centralized security monitoring.
- To connect and monitor endpoints using Wazuh Agents.
- To collect, analyze, and investigate Windows security logs and events.
- To identify successful and failed authentication activities using Event IDs 4624 and 4625.
- To simulate real-world attack scenarios such as brute-force attempts and malware detection.
- To monitor command execution activities and investigate generated security alerts.
- To develop security dashboards for visualizing alerts, agent status, and security events.
- To understand the workflow of a Security Operations Center and incident response process.
- To enhance practical skills in cybersecurity monitoring, threat hunting, and digital forensics.

This project helps bridge the gap between theoretical cybersecurity concepts and real-world SOC operations by providing hands-on experience with industry-relevant tools and techniques.

SOC ARCHITECTURE & LAB SETUP

To perform security monitoring and threat detection activities, a dedicated Security Operations Center (SOC) laboratory environment was established. The lab was designed to simulate a real-world enterprise monitoring environment where security events generated from endpoints could be collected, analysed, and investigated.

The environment was built using virtualization technology, allowing multiple systems to operate within an isolated and controlled infrastructure. Wazuh SIEM was deployed as the central monitoring platform responsible for collecting security logs, generating alerts, and providing visibility into endpoint activities.

A Windows endpoint was configured and connected to the Wazuh Manager through the Wazuh Agent. This configuration enabled centralized log collection and real-time security monitoring. The collected logs were used throughout the project for event analysis, threat detection, attack simulation, and incident investigation activities.

The SOC laboratory provided a practical environment for understanding how security analysts monitor systems, investigate alerts, and respond to potential threats using SIEM technologies.

Lab Components

Component	Purpose
Windows 10/11 Host Machine	Endpoint Monitoring
Wazuh Manager	SIEM & Log Collection
Wazuh Agent	Endpoint Monitoring
Sysmon	Advanced Event Logging
Kali Linux VM (Optional)	Attack Simulation
Wazuh Dashboard	Security Monitoring

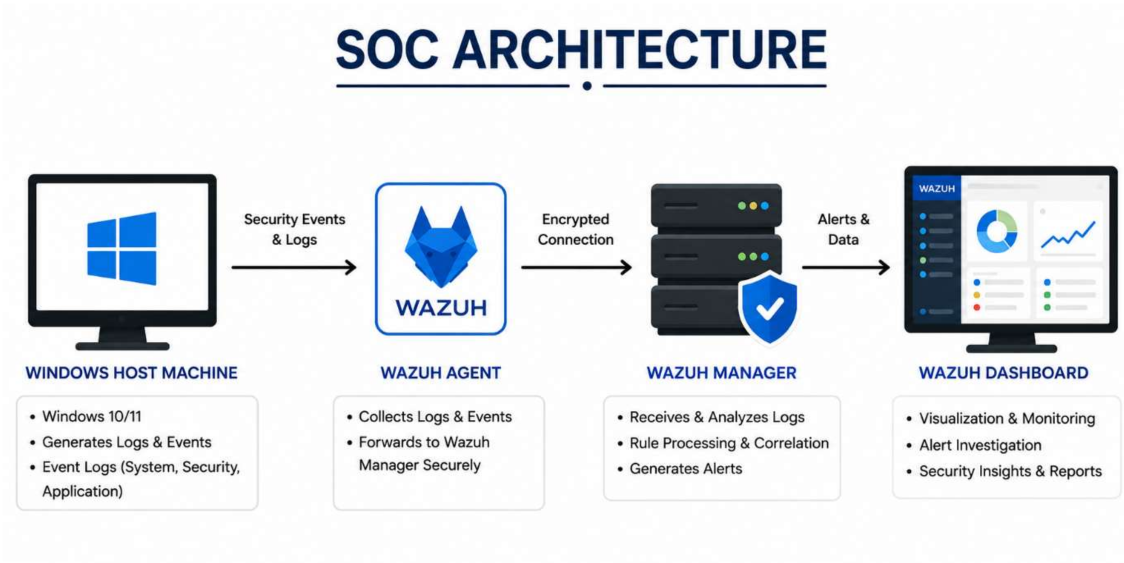

Figure 1: SOC Threat Detection Architecture Using Wazuh

The above architecture illustrates the Security Operations Center (SOC) monitoring environment implemented in this project. A Windows Host Machine acts as the endpoint and generates various system, security, and application logs. The Wazuh Agent installed on the endpoint collects these logs and securely forwards them to the Wazuh Manager. The Wazuh Manager performs log analysis, rule correlation, threat detection, and alert generation. Finally, all security events and alerts are visualized through the Wazuh Dashboard, enabling real-time monitoring, investigation, and incident response activities.

WAZUH SIEM DEPLOYMENT

Wazuh is an open-source Security Information and Event Management (SIEM) platform used for centralized log collection, security monitoring, threat detection, and incident response. In this project, Wazuh was deployed as the primary monitoring solution to collect and analyze security events generated from the Windows endpoint.

The deployment process involved configuring the Wazuh Manager and connecting the monitored endpoint through the Wazuh Agent. Once connected, the platform began collecting logs and generating alerts based on predefined security rules. This enabled continuous monitoring of system activities and improved visibility into potential security threats.

The Wazuh Dashboard served as the central interface for monitoring alerts, reviewing security events, and performing investigations. Through its visualization capabilities, analysts can quickly identify suspicious activities and respond to incidents efficiently.

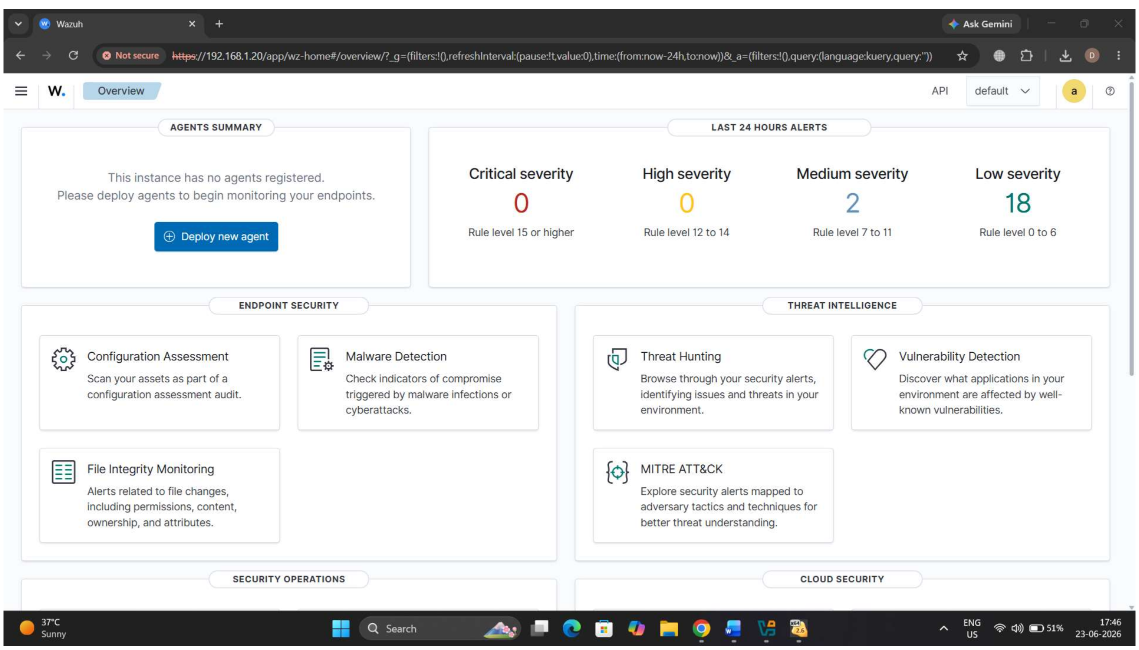

Figure 2: Wazuh Dashboard Successfully Deployed for Security Monitoring

The above figure shows the Wazuh Dashboard successfully deployed and accessible through the web interface. The dashboard provides centralized visibility into security events, threat intelligence modules, malware detection capabilities, file integrity monitoring, and endpoint security status. It serves as the primary platform for monitoring alerts, investigating incidents, and performing security analysis throughout the project lifecycle.

WAZUH AGENT CONFIGURATION

The Wazuh Agent is responsible for collecting logs and security-related events from monitored endpoints and forwarding them to the Wazuh Manager for analysis. After deploying the Wazuh platform, the next step was to install and configure the Wazuh Agent on the Windows host machine.

The agent continuously monitors system activities, Windows Event Logs, user authentication events, process creation events, and security-related activities. Once successfully connected to the Wazuh Manager, the endpoint becomes visible within the Wazuh Dashboard, allowing centralized monitoring and alert generation.

Proper agent registration is a critical step because all security events and detections depend on the successful communication between the endpoint and the Wazuh Manager.

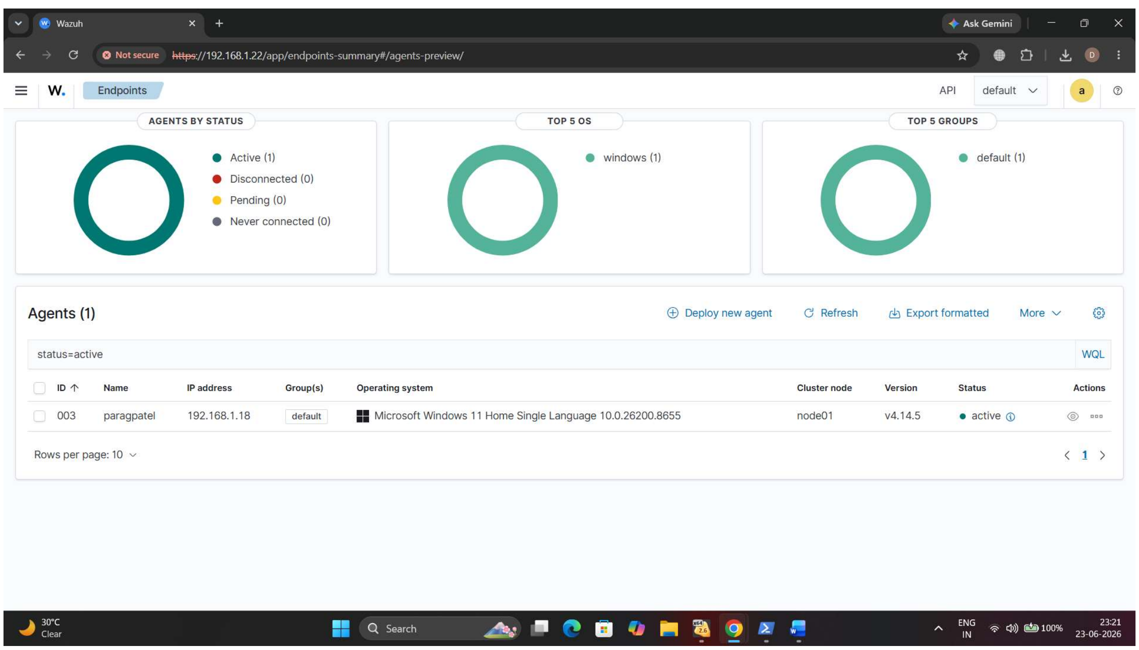

Figure 3 : Successfully Registered and Active Wazuh Agent

The above figure shows the successful registration and activation of the Windows endpoint within the Wazuh environment. The active status confirms that log collection and real-time security monitoring have been established successfully.

LOG COLLECTION & ANALYSIS

Log collection and analysis are essential components of Security Information and Event Management (SIEM). Wazuh continuously collects security logs from the monitored Windows endpoint and forwards them to the Wazuh Manager for centralized analysis.

For this project, Windows Event Logs were monitored to identify authentication activities, system events, and potential security incidents. Special attention was given to Event ID 4624 (Successful Login) and Event ID 4625 (Failed Login), as these events play a critical role in detecting unauthorized access attempts and brute force attacks.

The collected logs were analysed through both Windows Event Viewer and the Wazuh Dashboard to verify successful log ingestion and event correlation.

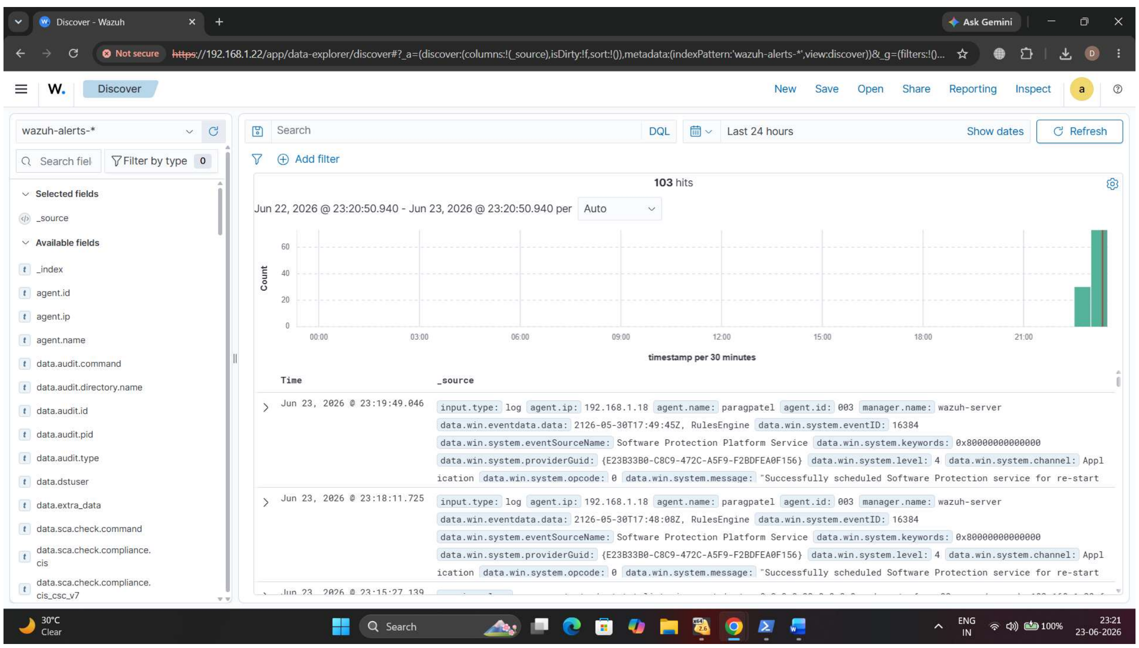

Figure 4: Windows Event Viewer Showing Security Logs

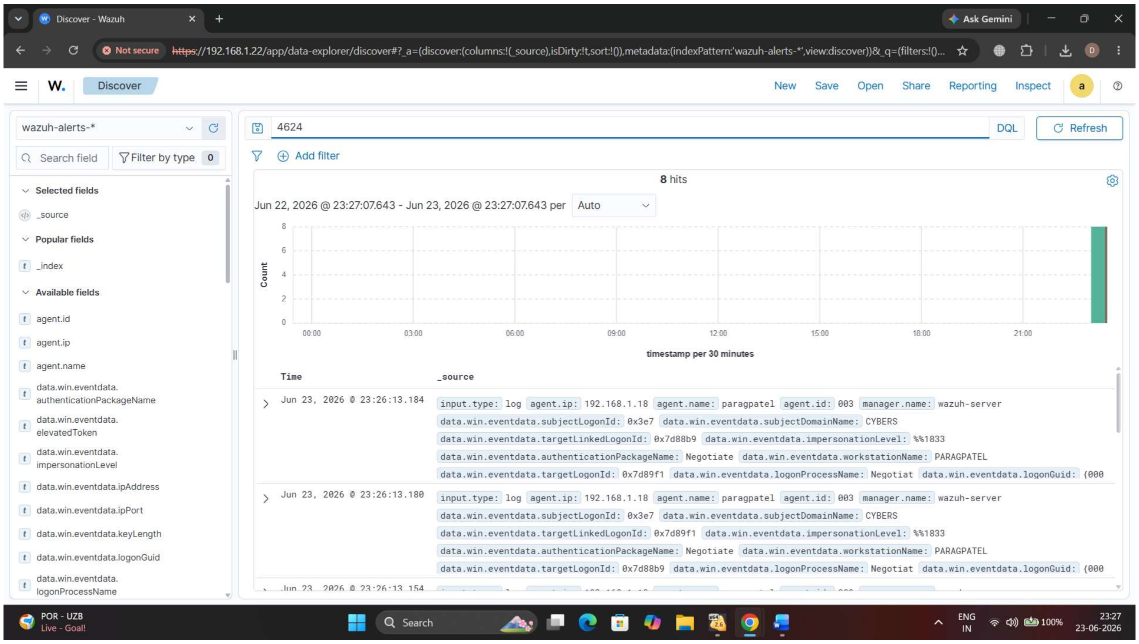

Figure 5: Event ID 4624 – Successful User Login Event

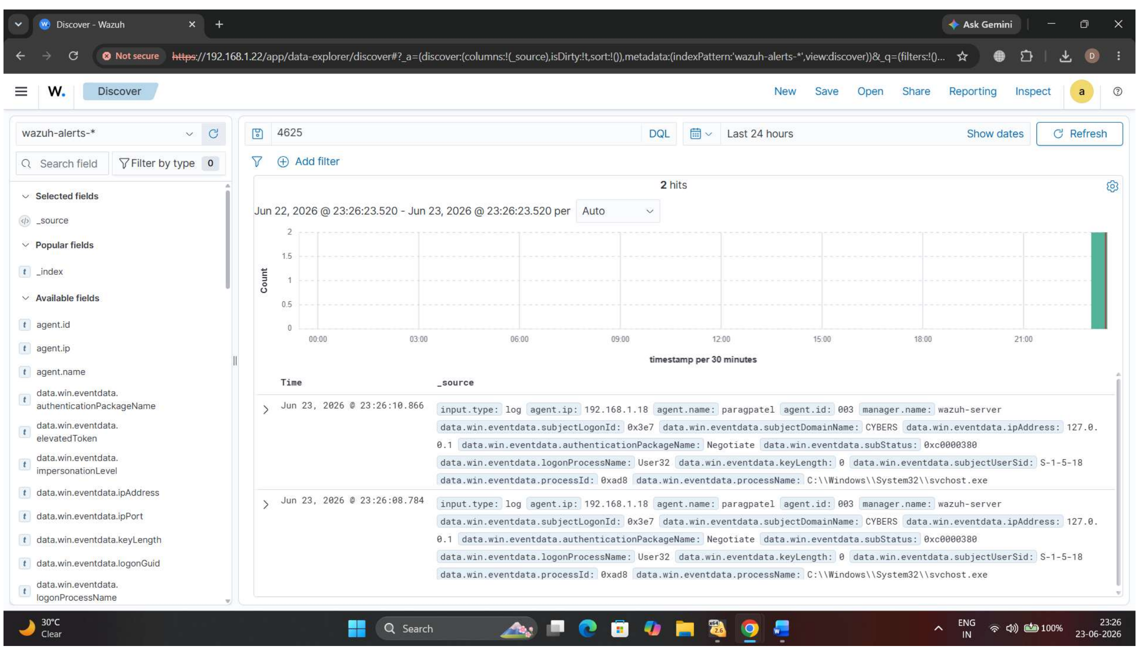

Figure 6: Event ID 4625 – Failed User Login Event

SECURITY EVENT MONITORING

Security Event Monitoring is one of the most important functions of a Security Operations Center (SOC). It enables security analysts to continuously observe system activities, identify suspicious behavior, and respond to potential threats before they impact organizational assets.

In this project, the Wazuh SIEM platform was used to monitor events generated from the Windows endpoint. The collected logs were automatically processed and categorized based on predefined security detection rules. Events such as user logins, failed authentication attempts, system activities, and process executions were monitored in real time.

The Wazuh Dashboard provided a centralized view of all detected events and alerts. This helped in understanding the security posture of the monitored system and identifying unusual activities that may require further investigation. Event monitoring also played a crucial role in validating log collection and ensuring that the deployed SIEM solution was functioning correctly.

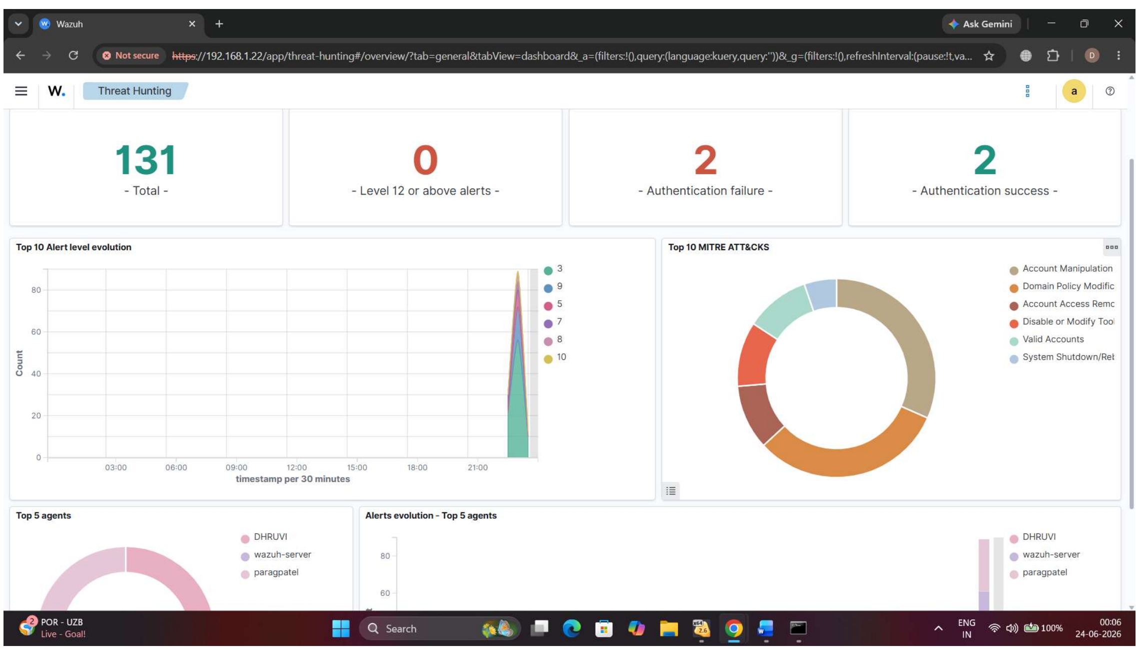

Figure 7: Top Security Events and Alerts Detected by Wazuh SIEM

The monitoring results confirmed that Wazuh successfully collected and analysed endpoint logs. Security events were categorized and displayed through the dashboard, providing real-time visibility into system activities and enabling efficient threat detection.

BRUTE FORCE ATTACK SIMULATION

Brute Force Attacks are among the most common techniques used by attackers to gain unauthorized access to systems. In this attack method, multiple login attempts are performed using incorrect credentials until valid authentication information is discovered.

To demonstrate Wazuh's detection capabilities, a controlled brute force simulation was performed on the Windows endpoint. Multiple failed login attempts were intentionally generated using invalid credentials. These authentication failures were recorded within Windows Security Logs and automatically forwarded to the Wazuh Manager for analysis.

The collected events triggered security alerts inside the Wazuh Dashboard, demonstrating the effectiveness of centralized monitoring and real-time threat detection. This simulation helped validate that Wazuh can successfully identify suspicious authentication activities and provide visibility into potential brute force attacks.

Attack Execution

The brute force simulation was performed by generating multiple failed authentication attempts on the Windows system. These failed login events were recorded as **Event ID 4625** within the Windows Security Log.

The generated events were then collected by the Wazuh Agent and analysed by the Wazuh Manager, resulting in security alerts visible through the dashboard.

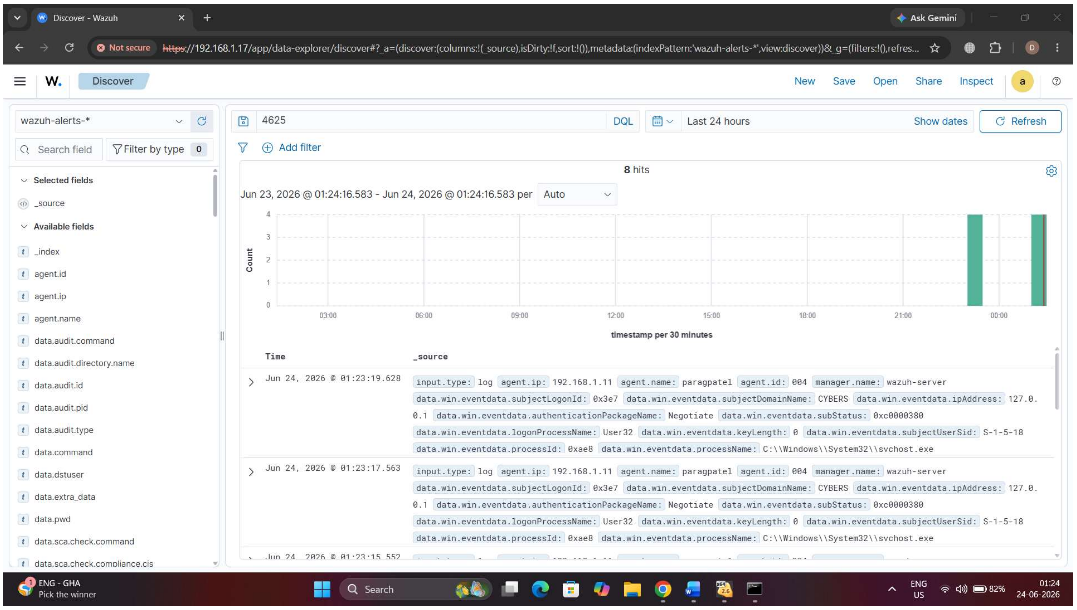

Figure 8: Failed Authentication Events Generated During Brute Force Simulation

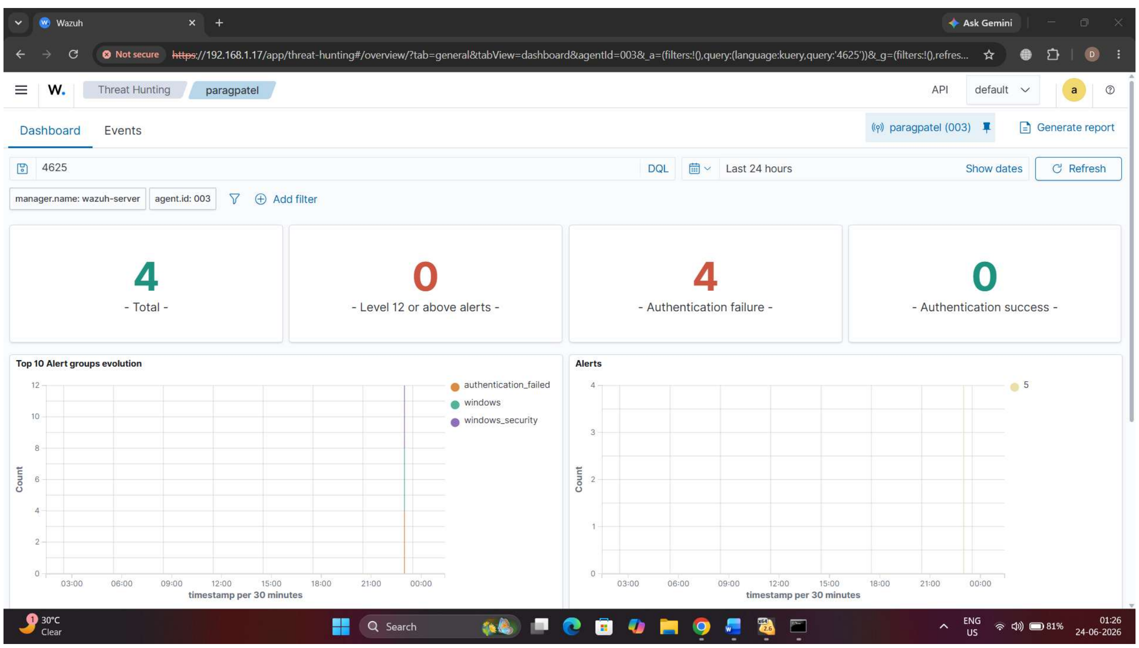

Figure 9: Wazuh Detection of Failed Authentication Attempts

COMMAND MONITORING & DETECTION

Command monitoring is an important aspect of endpoint security because attackers often use command-line utilities to perform reconnaissance, privilege escalation, and post-exploitation activities. Wazuh can monitor command execution events and generate alerts when suspicious activities are detected.

During this project, commands such as **whoami** and **net user** were executed on the monitored Windows endpoint. These commands were recorded by the operating system and collected through the Wazuh monitoring infrastructure.

The generated events were analysed through the Wazuh Dashboard to validate command monitoring capabilities and improve visibility into endpoint activities.

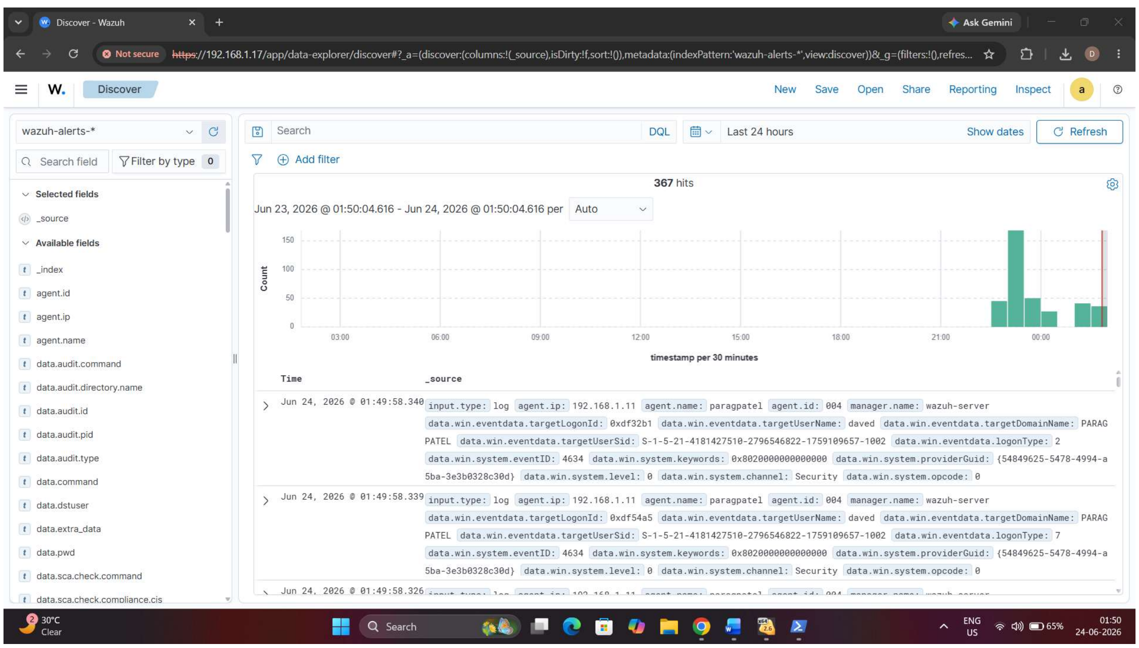

Figure 10: Command Execution Events Collected and Monitored in Wazuh SIEM

The above figure demonstrates command and system activity logs collected by the Wazuh Agent and analysed by the Wazuh Manager. The generated events provide visibility into endpoint activities and support security monitoring and threat investigation processes.

POWERSHELL ACTIVITY MONITORING

PowerShell is a powerful command-line and scripting framework used for Windows administration and automation. Due to its extensive capabilities, it is frequently used by both system administrators and threat actors. Therefore, monitoring PowerShell activities is essential for detecting suspicious behaviour and potential security threats.

During this project, PowerShell commands were executed on the monitored Windows endpoint to generate activity logs. These logs were collected by the Wazuh Agent and forwarded to the Wazuh Manager for analysis. The generated events provided visibility into administrative actions performed on the system.

The monitoring of PowerShell activities helps security analysts identify unauthorized command execution, suspicious scripts, and potential attack techniques used by adversaries.

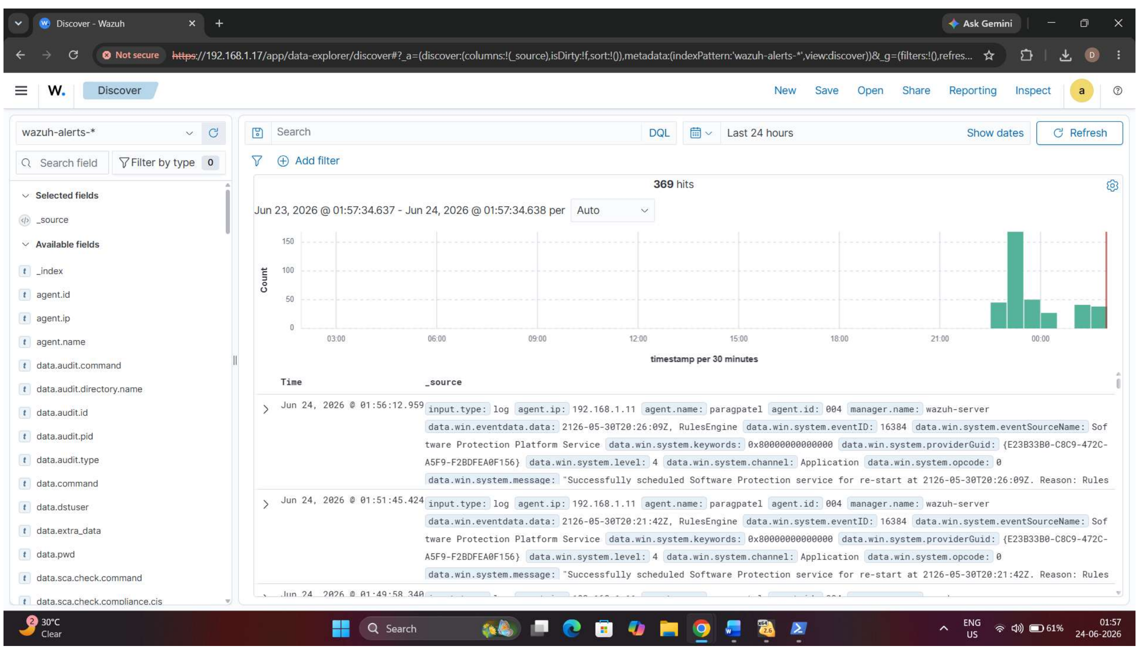

Figure 11: PowerShell Activity Events Detected and Monitored by Wazuh

The PowerShell monitoring process successfully captured command execution events from the endpoint. These logs improved visibility into administrative activities and demonstrated the effectiveness of Wazuh in monitoring Windows-based operations.

FINDINGS & OBSERVATIONS

The implementation of the Wazuh SIEM platform successfully demonstrated the process of centralized security monitoring and log analysis within a Windows environment. Throughout the project, multiple security events were generated, collected, and analyzed using the deployed monitoring infrastructure.

The following key findings were observed during the project:

- Wazuh successfully collected logs from the monitored Windows endpoint.
- The deployed Wazuh Agent maintained stable communication with the Wazuh Manager.
- Windows Security Events were continuously monitored and analyzed.
- Successful login events (Event ID 4624) were accurately recorded.
- Failed login events (Event ID 4625) were successfully detected and generated security alerts.
- Security event monitoring provided real-time visibility into endpoint activities.
- Brute force attack simulation generated detectable authentication failure events.
- Command execution activities were logged and monitored through the Wazuh platform.
- PowerShell activities were captured and analyzed for monitoring purposes.
- MITRE ATT&CK mapping provided additional threat intelligence context.
- Centralized dashboards improved incident visibility and investigation capabilities.

Overall Observation

The project demonstrated that Wazuh SIEM can effectively collect, analyze, and visualize security-related events. The generated alerts and monitoring dashboards provided valuable insights into endpoint activities and strengthened the overall security monitoring process.

RECOMMENDATIONS

Based on the observations and results obtained during the implementation of the Wazuh SIEM monitoring environment, several recommendations can be made to further improve security monitoring capabilities and strengthen the overall security posture of the organization.

Recommended Security Improvements

- Deploy Sysmon to enhance endpoint visibility and collect detailed process monitoring logs.
- Enable advanced Wazuh detection rules for identifying suspicious activities and attack patterns.
- Integrate Threat Intelligence feeds to improve detection of known malicious indicators.
- Implement regular security audits and log reviews to identify hidden threats.
- Configure email and real-time alert notifications for critical security incidents.
- Expand monitoring coverage by deploying Wazuh Agents across multiple endpoints.
- Utilize MITRE ATT&CK mapping for improved threat analysis and incident investigation.
- Conduct periodic vulnerability assessments and security testing.
- Maintain updated operating systems and security patches on all monitored devices.
- Develop a formal incident response process for handling detected security events.

Future Enhancements

Future improvements may include integrating additional security tools such as Sysmon, Suricata, Zeek, and Threat Intelligence platforms. These integrations can significantly improve detection accuracy, threat hunting capabilities, and overall security monitoring effectiveness.

Summary

By implementing the above recommendations, organizations can enhance their Security Operations Center (SOC) capabilities and improve their ability to detect, investigate, and respond to cyber threats efficiently.

CONCLUSION

This project successfully demonstrated the deployment and implementation of the Wazuh Security Information and Event Management (SIEM) platform for centralized security monitoring and threat detection. The environment was configured to collect, analyze, and visualize security events generated from a Windows endpoint.

Throughout the project, various monitoring activities were performed, including agent deployment, log collection, authentication monitoring, security event analysis, brute force attack simulation, command monitoring, and PowerShell activity monitoring. These activities provided practical experience in Security Operations Center (SOC) operations and incident monitoring.

The generated alerts and dashboards confirmed the effectiveness of Wazuh in identifying suspicious activities and providing real-time visibility into system events. The project also demonstrated the importance of centralized log management, threat detection, and security monitoring in modern cybersecurity environments.

Overall, this project enhanced practical knowledge of SIEM technologies, log analysis, threat hunting, and security operations while strengthening the technical skills required for careers in Cyber Security and Digital Forensics.

Final Outcome

The Wazuh SIEM platform was successfully deployed and configured, enabling centralized monitoring, alert generation, and security event analysis. The project objectives were achieved successfully, and the implemented environment demonstrated the core principles of modern SOC operations.

REFERENCES

The following resources were referred to during the implementation and completion of this project:

1. **Wazuh Official Documentation**
   - Wazuh Installation Guide
   - Wazuh Agent Configuration Guide
   - Wazuh Dashboard Documentation
2. **Microsoft Documentation**
   - Windows Event Viewer Documentation
   - Windows Security Auditing Events
   - PowerShell Documentation
3. **MITRE ATT&CK Framework**
   - Threat Detection Techniques
   - Adversary Tactics and Procedures
4. **OWASP Foundation**
   - Security Monitoring Best Practices
   - Cybersecurity Guidelines
5. **Cybersecurity and Infrastructure Security Agency (CISA)**
   - Security Monitoring Recommendations
   - Incident Response Guidelines
6. **Sysinternals Documentation**
   - Sysmon Configuration and Monitoring

7. **Security Information and Event Management (SIEM) Concepts**
    - Log Collection
    - Event Correlation
    - Alert Management
    - Threat Detection
8. **CFSS Summer Global Internship Resources**
    - Internship Learning Materials
    - Project Guidelines
    - SOC Operations & Threat Detection Modules

APPENDIX

Appendix A – Screenshots & Evidence

This appendix contains screenshots collected during the implementation and testing of the Wazuh SIEM environment. These screenshots serve as evidence of successful deployment, configuration, monitoring, threat detection, and analysis activities performed throughout the project.

Screenshots Included

Figure 1: SOC Threat Detection Architecture Using Wazuh

Figure 2: Wazuh Dashboard Successfully Deployed for Security Monitoring

Figure 3: Successfully Registered and Active Wazuh Agent

Figure 4: Windows Security Logs Collected in Wazuh SIEM

Figure 5: Event ID 4624 – Successful User Login Event

Figure 6: Event ID 4625 – Failed User Login Event

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Figure 11: PowerShell Activity Events Detected and Monitored by Wazuh